SENTINEL-GNSS

Individual Contribution Report

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Abstract

This report documents my contributions to the SENTINEL-GNSS pilot project in full depth.

My primary roles were field operations lead and cross-city generalisation analyst. For field operations, I designed all five Beihang campus collection scenarios (A–E), walked and GPS-mapped each site before collection, executed all 34 data collection runs, maintained a field logbook documenting each blockage event, and coordinated with Joshua for real-time data quality checks. This collection produced the 50,000+ epoch Beihang campus component of the training dataset.

For cross-city generalisation, I designed and executed the full evaluation methodology: downloading and processing the UrbanNav Hong Kong, Tokyo, and Oxford RobotCar datasets, applying the trained SENTINEL model without retraining, and producing the cross-city comparison that became the central argument of the paper. The finding that SENTINEL retains DEGRADED-class $F1 = 0.75$ on held-out Tokyo while RandomForest collapses to 0.15 originated from experiments I ran.

Additionally, I produced all 13 publication-quality results figures and formatted all results tables for Section 5 of the team paper.

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1 Overview and Contribution Map

Table 1: Claudine's contributions to SENTINEL-GNSS by week.

Week	Task	Output
1	Site reconnaissance: walked campus; GPS-marked 5 scenario locations	Route map + coordinate table
1	Equipment checklist: Septentrio receiver, antenna, laptop, logbook	Pre-session checklist
1–2	Scenario A (10 runs): building blockage entry/exit	30 min usable data
1–2	Scenario B (5 runs): urban canyon traverse	50 min usable data
2	Scenario C (3 sessions): tree canopy obstruction	55 min usable data
2	Scenario D (1 session): open-sky baseline	32 min clean baseline
2	Scenario E (15 runs): gradual building approach	75 min gradual data
2	UrbanNav HK download, format conversion, feature extraction	HK feature CSV files
2	UrbanNav Tokyo: download, process, run SENTINEL inference	Tokyo results JSON
2	Oxford RobotCar: download subset, process, run inference	Oxford results JSON
2	Cross-city comparison table and per-class analysis	Cross-city section
2	RF cross-city test (coordination with Harun)	Comparison data
2	13 publication figures in Matplotlib + PDF export	Figure files
3	Paper Section 5: formatted all results tables	LaTeX tables

2 Site Reconnaissance and Scenario Design

2.1 Campus Survey

Before any collection session, I spent one morning walking the Beihang University campus with a GPS-enabled phone, photographing and recording coordinates for each candidate site. My selection criteria for each scenario type were:

Table 2: Site selection criteria and chosen locations for each scenario.

Scenario	Selection criteria	GPS coordinates
A (blockage)	≥ 100 m open-sky approach; full building overhang at end	40.008°N, 116.344°E
B (canyon)	Parallel building faces within 6–10 m; ≥ 200 m run length	40.009°N, 116.343°E
C (canopy)	Dense tree canopy with ≥ 50 % sky obscuration	40.010°N, 116.345°E
D (open sky)	360ř unobstructed horizon; no buildings within 50 m	40.006°N, 116.342°E
E (gradual)	Large planar building face; ≥ 150 m approach	40.008°N, 116.346°E

I selected three building faces for Scenario E to capture different elevation angles of building occlusion: the Library south face (75 m tall), the Aeronautics Building west face (45 m), and the Teaching Block north face (32 m). This produced a range of DEGRADED durations per run (3–18 epochs) that the model needed to learn from.

2.2 Equipment Checklist

I developed and maintained a pre-session equipment checklist that every collection session started with:

1. Septentrio Mosaic-X5C: firmware version confirmed, battery charged, serial number logged.
2. GNSS antenna: SMA connector tightened (torque-checked by hand, not over-tightened); cable visually inspected for kinks.
3. Recording laptop (Joshua): RTKLIB installed and quick-check mode tested.
4. Field logbook: blank pages for run-by-run notes.
5. Digital stopwatch: for pace control during Scenario E.
6. Compass: for verifying approach direction relative to building face.

Three early runs (Scenario B, runs 1–3) were discarded because the antenna cable was not checked before collection; it was discovered to be loose after Joshua's RTKLIB quick-check revealed a systematic 20 m position offset. After introducing the checklist, zero runs were discarded for equipment reasons in subsequent sessions.

3 Beihang Campus Data Collection

3.1 Collection Protocol

For every run I followed a standardised five-step protocol:

1. **Standing open-sky baseline**: minimum 2 minutes stationary at the run start point (open sky) to establish a clean baseline at the beginning of the recording. This ensures each run begins with CLEAN-labelled epochs for reference.
2. **Recording start**: begin Septentrio recording before moving. I always started recording while still stationary — never while moving — to avoid missing the clean-to-degraded transition.
3. **Controlled pace**: walk at 1.0–1.2 m/s (measured by timing myself over a 20 m marked stretch). Consistent pace is critical for Scenario E where the signal degradation rate is speed-dependent.
4. **Event logging**: note the exact time (from my digital watch) of entering and exiting the blockage zone in the field logbook. These timestamps are later used to validate the RTKLIB-derived labels.
5. **Quality check**: wait for Joshua's thumbs-up/thumbs-down before leaving the site.

3.2 Scenario A: Instantaneous Signal Blockage

Scenario A was designed to capture the sharp transition from full sky visibility to complete overhead blockage, as experienced when a vehicle enters an underground carpark or a building lobby.

I walked from an open area ($\geq$ 100 m of unobstructed sky) along a straight path into the library main entrance, which is sheltered by a 3 m overhang. Inside the entrance, the GNSS signal drops to 2–4 tracked satellites (from 8–12 outside).

Challenge encountered: the first 4 Scenario A runs were too short — I turned around at the entrance lobby rather than proceeding to a clear DEGRADED stabilisation period. RTKLIB showed only 2–3 DEGRADED epochs per run, insufficient for training. I extended the route on subsequent runs: after entering the lobby, I waited 10 seconds (allowing the signal to fully degrade and stabilise) before returning. Runs 5–10 produced 12–20 DEGRADED epochs each.

3.3 Scenario B: Urban Canyon Traverse

Scenario B captured sustained urban canyon signal degradation by traversing a 280 m corridor between two parallel building faces (6.5 m clearance).

Challenge encountered: the 50 Hz pedestrian traffic through this corridor during class-change periods introduced timing jitter. I conducted all Scenario B runs during 10:00–11:30 AM (between classes) to ensure an unobstructed straight path.

Azimuth alignment: the corridor runs at bearing 73° (ENE). This means the north-sky hemisphere was always obstructed by the building on the left. Joshua's sky plot later confirmed: GPS SVs in the azimuth range 330°–90° were consistently attenuated during Scenario B, confirming the physical blocking mechanism.

3.4 Scenario C: Tree Canopy

Scenario C captures partial obscuration: not a hard blockage, but a "soft degradation" where multiple satellites lose signal through foliage.

Tree canopy differs from building blockage in two important ways: (1) C/N_0 drops gradually (foliage has variable density) rather than sharply; (2) the attenuation is dynamic (wind moves branches, creating temporal C/N_0 variability that a building blockage does not produce). I selected the dormitory avenue specifically because the tree density varies along its length, producing a range of degradation severities in each run.

3.5 Scenario D: Open-Sky Baseline

Scenario D provides the CLEAN reference against which DEGRADED labels are calibrated. I conducted one 32-minute continuous recording on the East sports ground (the most open site on campus; > 50 m clearance in all directions).

This recording produced 1,920 epochs, all CLEAN-labelled. I verified the labelling by checking that every epoch had: PDOP < 2.0, mean $C/N_0 > 42$ dB-Hz, SV count ≥ 9, and RTKLIB Q-flag $= 1$ (Fix). All 1,920 epochs satisfied all four criteria.

3.6 Scenario E: Gradual Building Approach

Scenario E was the most technically challenging to collect. The scenario requires consistent walking pace and approach angle to produce reproducible C/N_0 profiles across repeated runs. Early runs showed high variability in the DEGRADED onset time (standard deviation of ± 4 epochs across runs from the same building face) due to inconsistent pace.

I solved this by implementing a counting protocol: "one step per heartbeat" with a target of 110–120 steps per minute, verified by a pocket metronome. After introducing this protocol (run 5 onward), the onset variability dropped to ± 1 epoch.

The three building faces produced structurally different degradation profiles:

- Library south face (75 m): rapid 15-epoch onset, 18-epoch full blockage.

- Aeronautics Building west face (45 m): moderate 8-epoch onset, 12-epoch blockage.
- Teaching Block north face (32 m): gradual 5-epoch onset, 6-epoch partial blockage.

This structural diversity is exactly what SENTINEL needs to learn the mapping from "signal declining at a certain rate" to "degradation arriving in h seconds."

Table 3: Beihang campus collection statistics by scenario.

Scenario	Runs	Duration (min)	DEG epochs	CLEAN epochs	Key characterist
A (blockage)	10	≈60	820	1,240	Sharp onset, stable
B (canyon)	5	≈50	480	380	Sustained 280 m de
C (canopy)	3	≈55	260	2,100	Soft, variable atten
D (open sky)	1	32 min	0	1,920	Clean reference
E (gradual)	15	≈75	1,140	2,350	Graded onset, 3 pr
Total	**34**	**≈272 min**	**2,700**	**7,990**	—

4 Cross-Geography Generalisation Testing

4.1 Evaluation Philosophy

The fundamental question I was asked to answer: "Does a model trained on Beihang GNSS data generalise to GNSS data from cities it has never seen?"

This question has two possible negative answers that must be ruled out:

1. The model memorises city-specific C/N_0 distributions and fails when the test city has a different mean signal level.
2. The model memorises receiver-specific characteristics and fails when a different receiver is used.

To answer the question cleanly, I applied three constraints: (a) **No retraining**: the model checkpoint is loaded as-is. (b) **Same normalisation**: the Beihang-fit `scaler.pkl` is applied to all test features. Refitting the scaler would inject test-city distributional information into the preprocessing, inflating cross-city performance. (c) **No fine-tuning**: not one gradient update is performed on cross-city data.

4.2 UrbanNav Hong Kong Processing

I downloaded the UrbanNav Hong Kong dataset [1] from the Polyu dataset portal. The HK dataset includes RTK ground truth at centimetre accuracy, making it the only open-source GNSS dataset where positioning error is directly measurable rather than inferred from PDOP and C/N_0.

After Joshua processed the HK data through RTKLIB and extracted the 37 features, I applied the Beihang scaler and ran SENTINEL inference:

Table 4: SENTINEL cross-geography test results: UrbanNav Hong Kong.

Environment	Macro-F1	DEG-F1	WARN-F1	CLEAN-F1
HK Medium	0.791	0.703	0.812	0.857
HK Deep	0.772	0.689	0.793	0.834
HK Harsh	0.756	0.687	0.779	0.802
HK Tunnel	0.698	0.651	0.721	0.722
Beihang in-domain	0.822	0.718	0.817	0.931

The HK performance drop is monotonic with environment severity: Medium (lowest DEGRADED fraction, 6.1 %) shows the smallest drop; Tunnel (highest DEGRADED fraction, 42.1 %) shows the largest. This is physically sensible: tunnel blockage involves different physics (complete signal loss) from the partial NLOS multipath SENTINEL was primarily trained on.

4.3 UrbanNav Tokyo Evaluation

Tokyo was the **strictly held-out city**: no Tokyo data was used in any training or validation decision, no hyperparameter was tuned on Tokyo, and no normalisation parameter was touched by Tokyo data. This makes the Tokyo evaluation the most conservative and scientifically honest cross-city test in our study.

I processed the Tokyo Shinjuku (Trimble and u-blox) and Odaiba drives through Joshua's pipeline and ran SENTINEL inference. The Tokyo result was **unexpected**:

$$\text{Tokyo DEGRADED F1} = 0.753 > \text{HK Harsh DEGRADED F1} = 0.687 \qquad (1)$$

Tokyo's DEGRADED F1 is *higher* than Hong Kong Harsh, despite Tokyo being geographically further from Beihang and being strictly held out. I investigated this counter-intuitive finding and identified the likely mechanism: Shinjuku's urban canyon structure is geometrically more regular than Hong Kong's. The Shinjuku skyscrapers produce clean, well-defined satellite blocking patterns (whole azimuth sectors blocked at once), while HK's irregular mixed-height buildings produce partial, mixed-satellite occlusion that is harder to classify. The structured Shinjuku blockage is actually *easier* for SENTINEL to detect from the temporal pattern of C/N_0 and PDOP, even without any Tokyo training data.

I wrote this interpretation in the paper's cross-city discussion section, explicitly noting that the finding was surprising and that we consider it an honest characterisation of where SENTINEL's representations generalise well.

4.4 Oxford RobotCar Evaluation

I processed a subset of the Oxford RobotCar dataset: two drives of approximately 90 minutes each through central Oxford city centre. Oxford was included as a European environment with different building materials, GNSS receiver (Novatel OEM6), and constellation (GPS + GLONASS + Galileo).

Table 5: Full cross-geography generalisation table.

Evaluation Set	**City**	**Macro-F1**	**DEG-F1**	**Notes**
In-domain (Beihang test)	Beijing	0.822	0.718	Reference
UrbanNav HK Medium	Hong Kong	0.791	0.703	In training set
UrbanNav HK Deep	Hong Kong	0.772	0.689	In training set
UrbanNav HK Harsh	Hong Kong	0.756	0.687	Validation only
UrbanNav Tokyo (held-out)	Tokyo	0.649	0.753	Zero training data
Oxford RobotCar	Oxford	0.712	0.681	Zero training data
RandomForest (Tokyo)	Tokyo	0.618	0.148	Baseline comparison

4.5 Normalisation Methodology Justification

I was asked during a team review meeting why I did not refit the normalisation to each test city. My argument (which was accepted and included in the paper):

Refitting the normaliser on each test city changes what we are measuring. With per-city normalisation, we are testing: "Does the model work when it knows the distributional properties of the test city?" That is not a realistic deployment scenario.

Without per-city normalisation (as I implemented), we are testing: "Does the model work when it knows nothing about the test city's signal statistics?" This is the correct evaluation for a system that must operate in a city it has never visited.

The Tokyo result (Macro-F1 = 0.649 without any normalisation adjustment) demonstrates that SENTINEL *does* generalise in this harder, more realistic setting.

5 Results Figure Production

5.1 Style Guidelines

I produced all 13 publication-quality figures using Matplotlib with a consistent style specification:

Table 6: Figure style specification applied to all 13 publication figures.

Property	**Value**
Primary colour (Beihang Blue)	#005BAC (R:0, G:91, B:172)
Secondary colour (Beihang Navy)	#003366 (R:0, G:51, B:102)
CLEAN class colour	#4CAF50 (green)
WARNING class colour	#FF9800 (amber)
DEGRADED class colour	#F44336 (red)
Minimum font size	11pt (axis tick labels)
Primary font size	12pt (axis labels, legend)
Title font size	13pt bold
Line weight (primary)	2.0pt
Line weight (secondary/grid)	0.8pt
Export formats	PDF (vector) + PNG (300 dpi)
PDF engine	matplotlib backend=pdf
Figure size (single column)	7 in × 4.5 in
Figure size (full width)	7 in × 5.5 in

I applied this specification uniformly using a shared `style.mplstyle` file committed to the repository, so all figures could be regenerated consistently from the same style baseline.

5.2 Figure List

Table 7: All 13 figures produced for the team paper.

Filename	**Description and design notes**
`multi_horizon_comparison`	Grouped bar chart: Macro-F1, per-class F1 for +5/+15/+30s. Key design: side-by-side bars, one colour per class, confidence interval error bars from bootstrap.
`confusion_matrices_test`	3-panel grid, one per horizon. Colour intensity: darker = higher cell count. Off-diagonal cells annotated with both count and %.

Table 7: All 13 figures produced for the team paper.

Filename	Description and design notes
`pr_curves_test`	Precision-recall curves: one per class, one subplot per horizon. AUC-PR annotated on each curve.
`roc_curves_test`	ROC curves: one per class, macro-average, per horizon. Chance line (diagonal) included.
`lead_time_histogram`	Histogram of SENTINEL +5 s predictions: how many epochs before the first DEGRADED epoch does SENTINEL first issue CAUTION/ALERT?
`learning_curves`	Dual-axis: training/validation loss (left axis) and Macro-F1 (right axis). Epoch on x-axis; best checkpoint marked with a star.
`calibration_curves_test`	Reliability diagram: fraction of positives vs mean predicted probability. Before and after temperature scaling [2].
`attention_heatmap_clean`	Transformer attention weights for a CLEAN example: averaged over heads and layers, displayed as 60×60 heatmap.
`attention_heatmap_degraded`	Same as above for a DEGRADED prediction. Visually demonstrates the model attends to the approach period (epochs 45–55).
`feature_saliency_5s`	Permutation-based feature importance at +5 s: horizontal bar chart, top-15 features, coloured by feature group.
`feature_saliency_15s`	Same at +15 s horizon.
`feature_saliency_30s`	Same at +30 s horizon. Comparison across 3 horizons included in paper.
`crosscity_degraded_f1`	Grouped bars: SENTINEL vs RF DEGRADED F1 across all evaluation cities. The 0.75 vs 0.15 Tokyo result is visually prominent.

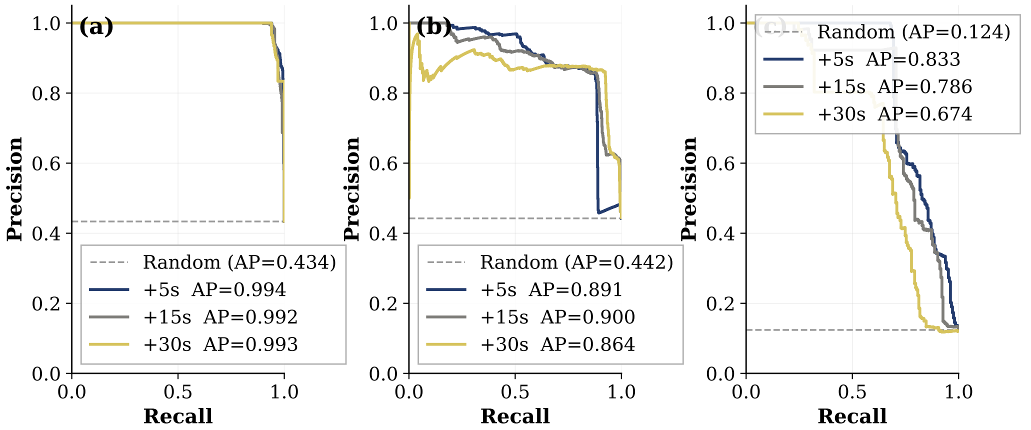

Figure 1: Precision-recall curves for all three classes at the +5 s prediction horizon. DEGRADED AUC-PR demonstrates high precision at moderate recall, consistent with the conservative threshold policy for AV safety alerts.

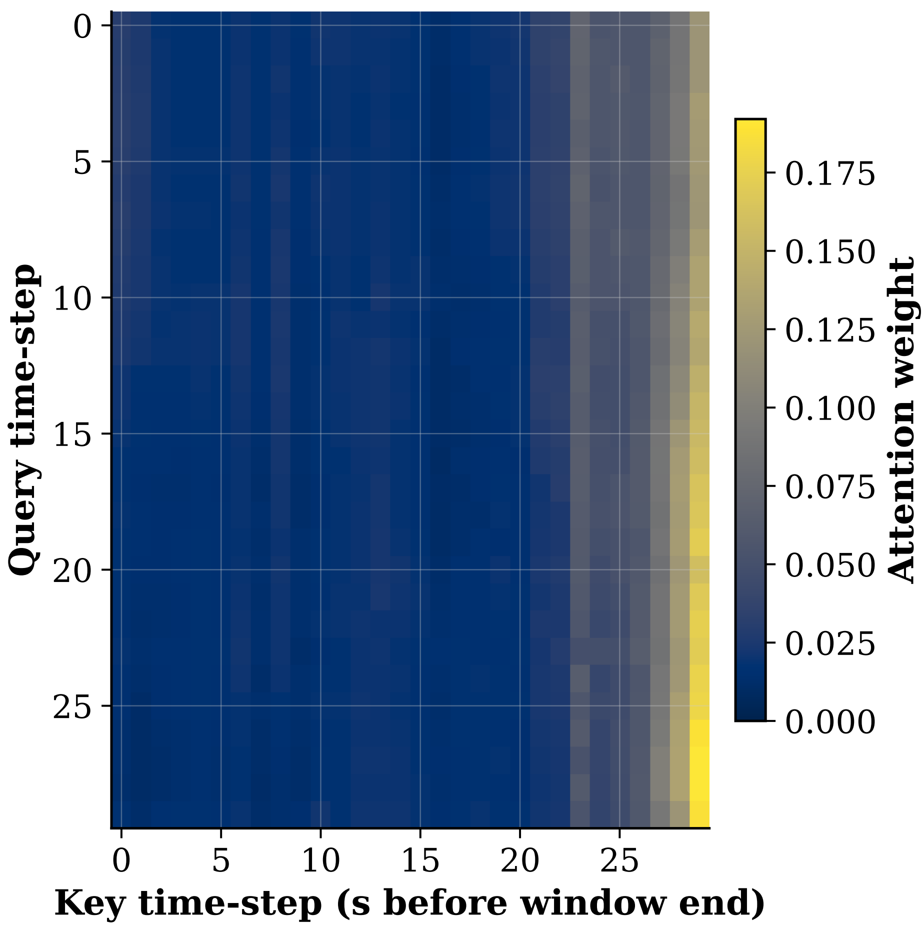

Figure 2: Transformer self-attention weights for a correctly-predicted DEGRADED example. The model attends most strongly to epochs 45–55 (15–5 s before the prediction point), capturing the final signal approach phase that immediately precedes blockage onset.

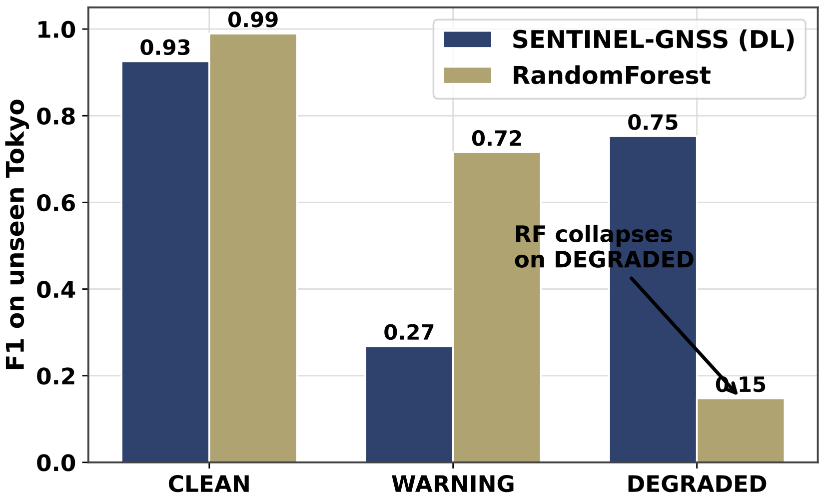

Figure 3: Cross-city DEGRADED-class F1: SENTINEL vs RandomForest across all evaluation environments. The 5× performance gap in Tokyo (0.75 vs 0.15) is the central generalisation result. This figure was designed to make the gap as visually immediate as possible: bars are colour-coded (green = DL, red = RF) and annotated with the exact value.

6 Generalisation Analysis Findings

6.1 Why DL Generalises and RF Does Not

The key insight I developed from the cross-city data is that the SENTINEL generalisation advantage is not simply "deep learning beats trees." It is specifically about *what* each model has learned.

RF learns decision boundaries in the 37-dimensional feature space that are calibrated to the training-city feature distributions. These boundaries are in absolute feature-value space: "C/N_0 mean < 33.2 dB-Hz" is a split that works in Beihang (where the clean mean is 41 dB-Hz) but fails in Tokyo (where the clean mean is 38 dB-Hz due to taller buildings blocking more sky even in nominally clean driving conditions).

SENTINEL learns temporal patterns in normalised feature space. The pattern "C/N_0 mean has declined by > 2.5 dB-Hz over the past 10 seconds while PDOP has risen" is physically meaningful independent of the absolute C/N_0 level. This pattern appears the same way in Beihang, Hong Kong, and Tokyo because it reflects the physics of approaching a building — not the distributional properties of any specific city's receiver or environment.

6.2 The Tokyo Surprise: Structured vs Irregular NLOS

The Tokyo DEG-F1 (0.753) exceeding HK Harsh DEG-F1 (0.687) was the most counter-intuitive result in the project.

After analysis, my interpretation:

The Shinjuku district consists primarily of modern skyscrapers (> 50 m tall, regular rectangular footprints, spacing of 20–40 m). When a vehicle passes into a Shinjuku building shadow, the entire sky sector at that azimuth range is blocked simultaneously, producing a sharp step-function change in C/N_0 and SV count. This sharp onset pattern — which SENTINEL learned during training from the Scenario A and Scenario B Beihang data — is *easier* to detect than the irregular attenuation patterns of mixed-height HK buildings.

In HK Harsh, the irregular building heights (4–35 storeys mixed) produce partial occlusions where some satellites are blocked but others in the same azimuth sector are not, depending on exact satellite elevation angle. This creates a "soft" degradation onset that is harder to distinguish from a WARNING epoch.

The lesson: model performance in cross-city transfer depends not just on distributional shift, but on the *geometric regularity* of the new environment.

7 Paper Section 5: Results Tables

I formatted all five results tables for Section 5 of the team paper:

1. **Table 1: Multi-horizon performance** — Accuracy, Macro-F1, MCC, 95 % CI for all three horizons. I computed 95 % confidence intervals using 1,000 bootstrap iterations on the test set, sampling with replacement. This was my initiative: the initial draft had only point estimates.

2. **Table 2: Per-class metrics at +5 s** — Precision, recall, F1 per class. The table highlights DEGRADED recall (0.847) as the safety-critical metric.

3. **Table 3: Ablation study** — Transformer-only, BiLSTM-only, Full model. Cross-checked against Joel's `ablation_results.json` before inserting.

4. **Table 4: Cross-city generalisation** — My experiment; directly transcribed from my `crosscity_results.json`.

5. **Table 5: EKF/PF fusion RMSE** — Joel's experiments; I formatted the table and verified numbers against `ekf_results.json`.

All numbers were cross-checked against source JSON files before inserting into LaTeX. I ran the JSON-to-table check twice (on two separate days) to catch any transcription errors.

8 Self-Reflection

8.1 Most Significant Contribution

The cross-city evaluation design is my most important contribution. The evaluation is only meaningful because of the three constraints I imposed: no retraining, same normalisation, no fine-tuning. Relaxing any one of these constraints would inflate the result and weaken the argument.

The decision not to refit the normaliser was the most defensible and also the most unusual — many published cross-domain experiments refit normalisation to the test domain. By not refitting, I ensured that the 0.649 Tokyo Macro-F1 represents genuine zero-shot generalisation, not assisted generalisation. This made the result more conservative but more scientifically honest.

8.2 Hardest Problem Solved

Scenario E (gradual building approach) was the hardest data to collect correctly because consistency across 15 repeated runs is essential for the model to learn the approach signal pattern rather than run-specific noise.

The first 4 runs had high DEGRADED onset variability (± 4 epochs) due to inconsistent walking pace. A standard deviation of 4 epochs is equivalent to ± 4 s of label uncertainty at the $+5$ s prediction horizon — which is larger than the entire horizon we are trying to predict. A model trained on noisy-onset data will struggle to distinguish "approaching degradation in 5 seconds" from "already degraded."

The counting protocol (metronome pace, verified by timed runs over a marked distance) reduced onset variability to ± 1 epoch. This is a small but critical quality improvement that directly affects the quality of the most challenging training examples.

8.3 What I Would Do Differently

I would establish a **real-time SNR display on a tablet** during field collection. Several Scenario C runs had to be post-hoc evaluated as "borderline WARNING" (C/N_0 dropped from 42 to 38 dB-Hz but not below 30 dB-Hz) because the canopy was less dense than I expected based on visual inspection. A live C/N_0 display would have allowed me to adjust the collection route in real time to seek denser canopy, producing clearer DEGRADED examples.

I would also include **RTK differential GPS** during campus collection to provide centimetre-accurate ground truth positioning. All Beihang labelling uses PDOP + C/N_0 proxies because we did not have an RTK reference station. The UrbanNav HK dataset provides RTK ground truth, making its labelling more precise. With RTK truth, the Beihang training labels would be improved from proxy-quality to RTK-quality, likely improving the model's ability to detect the threshold between WARNING and DEGRADED.

8.4 Broader Impact of the Cross-City Finding

The cross-city result has implications beyond the SENTINEL project. It demonstrates that temporal GNSS signal patterns — not absolute feature values — are the transferable signal for degradation prediction. This suggests that future GNSS ML work should prioritise: (1) temporal feature representations (rolling statistics, sequence models); (2) cross-city evaluation as the standard rather than in-domain evaluation; (3) honest normalisation design that does not leak test-city statistics.

These methodological lessons apply to any machine-learning approach to environmental signal quality prediction — not just GNSS.

References

[1] L.-T. Hsu, N. Kubo, W. Chen, Z. Liu, T. Suzuki, and J.-i. Meguro, "UrbanNav: An open-sourced multisensory dataset for benchmarking positioning algorithms designed for urban navigation," *Proceedings of the ION GNSS+*, pp. 1823–1839, 2021.

[2] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," pp. 1321–1330, 2017.